Q. Implement following:

A. WAP to operation-creating, accessing, slicing, modifying, adding, removing, sorting,reversing, finding lengths, and looping through lists.

B. WAP to get user id, user name, and user age from user and based on the entered id print the details for particular user. Hint: use dictionary.

1. Various list Functions:

1. **Creating:** We create a list myList with some initial values.
2. **Accessing:** Use indexing to access the first and last elements.
3. **Slicing:** Extract a part of the list using slicing.
4. **Modifying:** Change the value of an element using its index.
5. **Adding:** Use append and extend to add single and multiple elements.
6. **Inserting:** Insert an element at a specific index using insert.
7. **Removing:** Remove a specific element using remove and use pop to remove the lastelement.
8. **Sorting:** Use sort to sort the list in ascending order.
9. **Reversing:** Use reverse to reverse the list.
10. **Finding length:** Use len() to find the length of the list.
11. **Looping:** Loop through the list using a for loop to print each element. 12. **Index:** Finds the index of the first occurrence of an element in the list. 13. **Count**: Counts how many times a specific element appears in the list.
14. **Copy:** Creates a shallow copy of the list, meaning the new list has the same elements, butthey are stored at different memory locations.
15. **Clear:** Removes all elements from the list, making it an empty list.

2. Code for the Experiment:

A.)
```
my_list = [10, 20, 30, 40, 50]
print(my_list)
print( my_list[-1])

print("last element", my_list[-1])
print("first 3 elements", my_list[:3])
print("last 2 elements", my_list[-2:])

my_list[2] = 35
print(" after modification", my_list)

my_list.append(60)
print(" after appending 60", my_list)

my_list.insert(1, 15)
print(" after inserting 15 at index 1", my_list)

my_list.remove(35)
print(" after removing 35", my_list)

popped_element = my_list.pop(2)
print("element at index 2", my_list)

my_list.sort()
print("Sorted List", my_list)

my_list.reverse()
print("Reversed List", my_list)

print("Length of the List", len(my_list))

print("Looping through the list")
for item in my_list:
    print(item)
```

```
user_data = {}

# Number of users to input
n = int(input("Enter number of users: "))

# Get user id, name, and age from the user and store them in the dictionary
for i in range(n):
    user_id = input(f"\nEnter User {i+1} ID: ")
    user_name = input(f"Enter User {i+1} Name: ")
    user_age = int(input(f"Enter User {i+1} Age: "))

    # Store the user details in the dictionary
    user_data[user_id] = {"Name": user_name, "Age": user_age}

# Get the ID from the user to display details
search_id = input("\nEnter User ID to search: ")

# Display the user details if ID is found in the dictionary
if search_id in user_data:
    print("\nUser Details:")
    print(f"ID: {search_id}")
    print(f"Name: {user_data[search_id]['Name']}")
    print(f"Age: {user_data[search_id]['Age']}")
else:
    print(f"User with ID {search_id} not found.")
```

3. Result/Output/Writing Summary:

Output for List

```
Output                                                    Clear
[10, 20, 30, 40, 50]
50
last element 50
first 3 elements [10, 20, 30]
last 2 elements [40, 50]
 after modification [10, 20, 35, 40, 50]
 after appending 60 [10, 20, 35, 40, 50, 60]
 after inserting 15 at index 1 [10, 15, 20, 35, 40, 50, 60]
 after removing 35 [10, 15, 20, 40, 50, 60]
element at index 2 [10, 15, 40, 50, 60]
Sorted List [10, 15, 40, 50, 60]
Reversed List [60, 50, 40, 15, 10]
Length of the List 5
Looping through the list
60
50
40
15
10

=== Code Execution Successful ===
```

Output for Dictionary

```
Output                                                    Clear
Enter number of users: 3

Enter User 1 ID: 1
Enter User 1 Name: abc
Enter User 1 Age: 12

Enter User 2 ID: 2
Enter User 2 Name: def
Enter User 2 Age: 11

Enter User 3 ID: 3
Enter User 3 Name: ghi
Enter User 3 Age: 13

Enter User ID to search: 3

User Details:
ID: 3
Name: ghi
Age: 13
```

Learning outcomes (What I have learnt):

1. Learned about List in python.
2. Learned about various functions of list.
3. Learned about Dictionary.